

Combined Directional Support Transfer

Rayleigh Forcing, Frame Volume, Leading Scale, and Capacity in One Bound

Prime Dynamics Theory Program

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Abstract

We combine RH-120, RH-123, and RH-124 into one directional support theorem. Let $V = \sqrt{\det G}$ and $B = (1 - \gamma)_+^4 V / (L^4 C)$. If the gauged next level has retained Gram $a(1 - \eta)$, tail parameters b, δ , determinant $|\det S|$, leading factor ℓ , and capacity factor c , then with

$$\hat{\gamma}'^2 = \frac{b\gamma^2 + \delta}{a(1 - \eta)}$$

we have

$$B' \geq \frac{[a(1 - \eta)]^2 |\det S|}{\ell^4 c} (1 - \hat{\gamma}')_+^4 \frac{V}{L^4 C}.$$

The formula is sharp when every input inequality is an equality. On the explicitly conditioned RH-121 archive, all 96 one-step transfers are nonzero and all 24 five-scale chains finish above 10^{-8} . The audit is a regularized consistency test, not a physical or asymptotic support proof.

1 The combined candidate

The RH-114 directional lower, after RH-110 capacity division, is

$$B(G, D; L, C) = \frac{(1 - \gamma(G, D))_+^4 \sqrt{\det G}}{L^4 C}.$$

All factors must refer to the same enclosed operator. Treating them as independent black boxes can lose the correlation quantified in RH-124.

Theorem 1 (Combined directional transfer). *Assume the RH-123 hypotheses*

$$G' \succeq a(1 - \eta)S^*GS, \text{quad} D' \preceq bS^*DS + \delta S^*GS,$$

and suppose $L' \leq \ell L$ and $C' \leq cC$. Define $\hat{\gamma}'^2 = (b\gamma^2 + \delta)/[a(1 - \eta)]$. In four dimensions,

$$B' \geq \frac{[a(1 - \eta)]^2 |\det S|}{\ell^4 c} (1 - \hat{\gamma}')_+^4 \frac{\sqrt{\det G}}{L^4 C}.$$

Proof. RH-123 gives $\gamma' \leq \hat{\gamma}'$. Determinant monotonicity gives $\sqrt{\det G'} \geq [a(1-\eta)]^2 |\det S| \sqrt{\det G}$. Insert these three one-sided inequalities into the definition of B' . $\square$

These are standard positive-matrix and singular-value operations [Bhatia, 2007, Horn and Johnson, 1991]. Equality is attained by scalar relative tails, exact Gram scaling, and exact choices $L' = \ell L$, $C' = cC$.

Corollary 2 (Iterated lower). *If the one-step theorem gives $B_{n+1} \geq m_n B_n$, then $B_N \geq B_0 \prod_{n < N} m_n$. Any independently established lower for B_0 may replace the exact source value.*

2 Five-scale conditioned audit

We apply the exact-Gram gauges from RH-121 to 96 adjacent-scale pairs. Thus $a = 1$, $\eta = \delta = 0$, $b = b_{\text{opt}}$, while ℓ and c are the observed ratios of the archived leading and capacity uppers. Every target lower is compared with the target candidate computed on the same regularized Gram path.

There are zero one-step dominance failures. All 96 transfers are nonzero; 95 exceed 10^{-8} . Grouping by side, threshold, and normalized phase gives 24 chains. Their terminal lowers range from 1.13×10^{-7} to 8.27×10^{-2} , with median 2.59×10^{-2} . All 24 exceed 10^{-8} , 23 exceed 10^{-6} , and 21 exceed 10^{-4} .

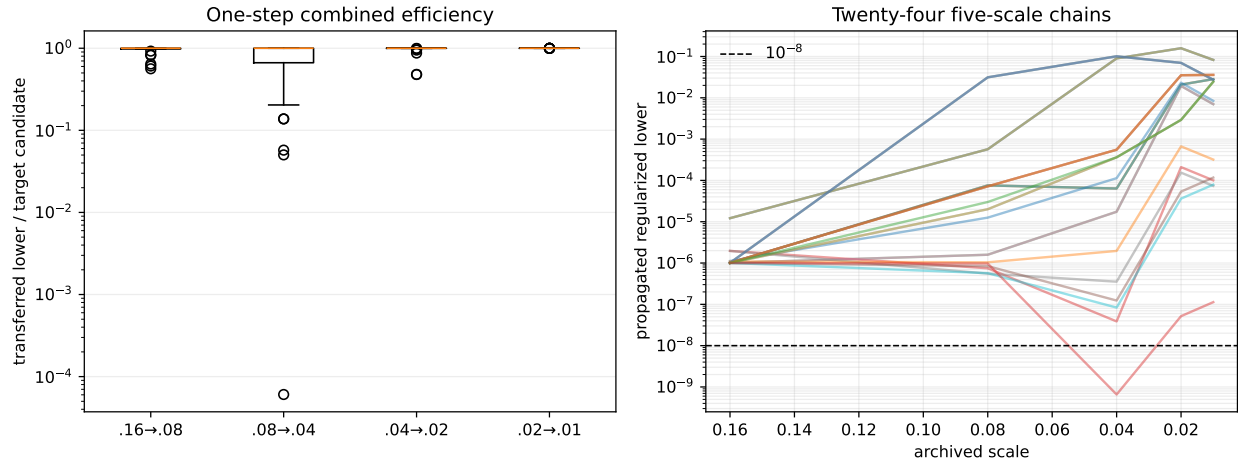


Figure 1: One-step efficiency and iterated regularized lower chains.

The good finite propagation does not erase the RH-121 boundary. Its 10^{-12} Gram conditioning floor is not a physical lower enclosure, so these chains cannot be promoted to support certificates. RH-126 develops the independent direct-margin fallback; RH-127 supplies outward guards. No uniform Stage A, Hilbert–Polya operator, zeta-zero identification, or Riemann Hypothesis conclusion is claimed.

References

Rajendra Bhatia. *Positive Definite Matrices*. Princeton University Press, 2007.

Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991.